Putnam

Generating functions

Written by a model for Adit. Not Adit's words.

2026-11-07

This is a part of the full document `putnam.pdf`; the summary with the marks is `content/teach/putnam.md`. Problems are named by source and number only; nothing is reproduced. Written by a model for Adit; not Adit's words.

1 Generating functions

Recognise it. The nth term of a sequence depends on sums of products of earlier terms, which is a convolution. A count splits an object into an ordered sequence of parts, each part counted by a known sequence. A recurrence has coefficients that are polynomials in n. A sum of binomial coefficients runs over every third index, or over an arithmetic progression of indices. A problem asks about the coefficients of a power of a polynomial. The seminar's eleventh set accepts only this method, and its problems are the ladder.

Definition 1.1. The ordinary generating function of a sequence $(a_n)_{n\geq 0}$ is the formal power series $A(x) = \sum_{n\geq 0} a_n x^n$; the exponential generating function is $\hat{A}(x) = \sum_{n\geq 0} a_n x^n/n!$. Formal series are added and multiplied coefficientwise and by the Cauchy product; a series with nonzero constant term has a reciprocal; no convergence is involved until a number is substituted for x. We write $[x^n]A$ for the coefficient of x^n.

Theorem 1.2 (Products are convolutions)**.** *For ordinary and for exponential generating functions respectively,*

$$[x^n]\,A(x)B(x) = \sum_{k=0}^{n} a_k b_{n-k}, \qquad [x^n/n!]\,\hat{A}(x)\hat{B}(x) = \sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k}.$$

Hence: if an object of size n is an ordered pair of a first part of size k counted by a_k and a second of size $n-k$ counted by b_{n-k}, the objects are counted by AB; if the parts are built on labelled elements and the labels are distributed between the parts, the objects are counted by $\hat{A}\hat{B}$. An ordered sequence of any number of parts, each counted by C with $c_0 = 0$, is counted by $1/(1-C(x))$.

Proof. Multiply out: $x^k \cdot x^{n-k} = x^n$. For the exponential version, $\frac{x^k}{k!}\cdot\frac{x^{n-k}}{(n-k)!} = \binom{n}{k}\frac{x^n}{n!}$, and $\binom{n}{k}$ is the number of ways to choose which labels go to the first part. The sequence statement is $\sum_{j\geq 0} C(x)^j$, a geometric series that makes sense formally because $c_0 = 0$ means C^j starts at x^j and each coefficient is a finite sum. □

Theorem 1.3 (Linear recurrences and rational functions)**.** *A sequence satisfies $a_{n+k} = c_{k-1}a_{n+k-1} + \cdots + c_0 a_n$ for all $n \geq 0$ if and only if $Q(x)A(x)$ is a polynomial of degree less than k, where $Q(x) = 1 - c_{k-1}x - \cdots - c_0 x^k$ is the reversed characteristic polynomial. Partial fractions then give the closed form: if $Q(x) = \prod_i (1 - r_i x)^{m_i}$ then $A(x) = \sum_i \sum_{j=1}^{m_i} \dfrac{\alpha_{ij}}{(1 - r_i x)^j}$ and*

$$[x^n]\frac{1}{(1-rx)^j} = \binom{n+j-1}{j-1} r^n,$$

a polynomial in n of degree $j-1$ times r^n.

Proof. The coefficient of x^{n+k} in $Q(x)A(x)$ is $a_{n+k} - c_{k-1}a_{n+k-1} - \cdots - c_0 a_n$, which vanishes for all $n \geq 0$ exactly when the recurrence holds; so QA has no terms beyond x^{k-1}. The coefficient formula is the binomial series $(1-y)^{-j} = \sum_n \binom{n+j-1}{j-1} y^n$, proved by differentiating the geometric series $j-1$ times, with $y = rx$. □

A recurrence with polynomial coefficients, like $a_{n+1} = (n+1)a_n + \ldots$, becomes a differential equation for A, since $xA'(x) = \sum n a_n x^n$; that is when the exponential generating function usually gives the simpler equation.

Theorem 1.4 (Roots-of-unity filter)**.** *Let $\omega = e^{2\pi i/k}$ and let $A(x)$ be a polynomial, or a series converging at the points ω^j. Then for any residue r,*

$$\sum_{n \equiv r \pmod{k}} a_n = \frac{1}{k} \sum_{j=0}^{k-1} \omega^{-jr} A(\omega^j).$$

Proof. $\sum_{j=0}^{k-1} \omega^{j(n-r)}$ is k when $k \mid n-r$ and otherwise a geometric sum $\frac{\omega^{k(n-r)}-1}{\omega^{n-r}-1} = 0$. Multiply a_n by this, sum over n, and swap the finite sum over j with the sum over n. □

For $k = 2$ this is $\sum_{n \text{ even}} a_n = \frac{A(1)+A(-1)}{2}$. For $k = 3$ with $\omega^2 + \omega + 1 = 0$ the values $A(\omega)$ and $A(\omega^2)$ are conjugate when A has real coefficients, so the filter is $\frac{1}{3}\big(A(1) + 2\,\mathrm{Re}\,\omega^{-r}A(\omega)\big)$.

Lemma 1.5 (Standard series)**.** $\dfrac{1}{1-x} = \sum x^n$; $\dfrac{1}{(1-x)^m} = \sum_n \binom{n+m-1}{m-1} x^n$; $(1+x)^n = \sum_k \binom{n}{k} x^k$; $e^x = \sum x^n/n!$ *is the exponential generating function of the all-ones sequence, and $e^x - 1$ of the sequence that is 0 at $n = 0$ and 1 after;* $\dfrac{1 - \sqrt{1-4x}}{2x}$ *generates the Catalan numbers;* $\dfrac{x}{1 - x - x^2}$ *generates the Fibonacci numbers with $F_0 = 0$, $F_1 = 1$.*

Proof. The first three are the geometric and binomial series. The exponential ones are the definition. The Fibonacci one is the previous theorem with $Q = 1 - x - x^2$ and $QA = x$. For Catalan, the recurrence $C_{n+1} = \sum_k C_k C_{n-k}$ says $C(x) - 1 = xC(x)^2$, and the quadratic formula gives the root with $C(0) = 1$. □

First moves. Write the series. Multiply the recurrence by x^{n+k} and sum over n; collect $A(x)$ and the boundary terms. Decompose the object into parts and multiply the parts' series. Find the closed form and do partial fractions. For every kth coefficient, evaluate at the kth roots of unity. For labelled objects, use exponential generating functions. When a closed form is not available, extract what is asked (a sign, a residue, a growth rate) directly from the functional equation for A.

Example 1.6. Call a string over the alphabet $\{A, B, C\}$ tidy if every maximal run of consecutive A's has even length. Let t_n be the number of tidy strings of length n. Find a recurrence for t_n and a closed form.

Solution. A tidy string is an alternation: a run of A's of even length (possibly empty), then a letter B or C, then an even run of A's, then a letter B or C, and so on, ending with an even run. Even runs are counted by $E(x) = 1 + x^2 + x^4 + \cdots = 1/(1-x^2)$, and a letter B or C by $2x$. A tidy string is an even run followed by an ordered sequence of blocks, each block a letter followed by an even run, so by the convolution theorem

$$T(x) = E(x) \cdot \frac{1}{1 - 2xE(x)} = \frac{1}{1-x^2} \cdot \frac{1}{1 - \frac{2x}{1-x^2}} = \frac{1}{1 - 2x - x^2}.$$

Reading the denominator as a reversed characteristic polynomial, $t_n = 2t_{n-1} + t_{n-2}$ for $n \geq 2$, with $t_0 = 1$ and $t_1 = 2$. Check: $t_2 = 5$, and the tidy strings of length 2 are AA and the four strings with no A.

The characteristic roots are $1 \pm \sqrt{2}$, so $t_n = \alpha(1+\sqrt{2})^n + \beta(1-\sqrt{2})^n$; from $t_0 = 1$ and $t_1 = 2$,

$$t_n = \frac{(1+\sqrt{2})^{n+1} - (1-\sqrt{2})^{n+1}}{2\sqrt{2}}.$$

At $n = 0$ this is $2\sqrt{2}/(2\sqrt{2}) = 1$ and at $n = 1$ it is $4\sqrt{2}/(2\sqrt{2}) = 2$. The recurrence has a direct explanation too: a tidy string ends in B or C preceded by a tidy string of length $n-1$, or in AA preceded by a tidy string of length $n-2$; the generating function found it without that insight, and it also proves the decomposition is exhaustive and unambiguous. □

Example 1.7. Let $n \geq 1$. Find the number of pairs (A, B) of subsets of $\{1, \ldots, n\}$ with $A \subseteq B$ and $|B \setminus A|$ divisible by 3.

Solution. As in the counting section, each element is outside B, in $B \setminus A$, or in A, and marking the middle state with x gives the polynomial $(2+x)^n = \sum_k \binom{n}{k} 2^{n-k} x^k$, whose coefficient of x^k counts pairs with $|B \setminus A| = k$. The filter with $k = 3$, $r = 0$ and $\omega = e^{2\pi i/3}$ gives

$$N = \frac{1}{3}\Big(3^n + (2+\omega)^n + (2+\omega^2)^n\Big).$$

Now $2 + \omega = \frac{3}{2} + \frac{\sqrt{3}}{2}i = \sqrt{3}\, e^{i\pi/6}$ and $2 + \omega^2$ is its conjugate, so

$$N = \frac{1}{3}\Big(3^n + 2 \cdot 3^{n/2} \cos\frac{n\pi}{6}\Big).$$

Check at $n = 1$: $\frac{1}{3}(3 + 2\sqrt{3} \cdot \frac{\sqrt{3}}{2}) = 2$, the pairs $(\emptyset, \emptyset)$ and $(\{1\}, \{1\})$. At $n = 2$: $\frac{1}{3}(9 + 2 \cdot 3 \cdot \frac{1}{2}) = 4$, the four pairs with $A = B$, since $|B \setminus A| = 3$ is impossible and $|B \setminus A| \in \{1, 2\}$ is excluded. At $n = 3$: $\frac{1}{3}(27 + 0) = 9$, which is the 8 pairs with $A = B$ plus $(\emptyset, \{1, 2, 3\})$.

The parity version, $|B \setminus A|$ odd, is the same filter with $k = 2$ and gives $\frac{1}{2}(3^n - 1)$; the involution in the counting section proved that without complex numbers, and no such involution is in sight for divisibility by 3, which is what the roots of unity are for. □

Trap. Convergence is irrelevant for a formal power series and relevant the moment a number is substituted: the roots-of-unity filter on an infinite series needs the series to converge at those points, and a closed form like $1/(1-2x)$ represents the sequence 2^n for $|x| < \frac{1}{2}$ and nothing at $x = 1$. A recurrence that starts late, so the first few

coefficients need their own correction terms in QA; always compute QA to the order x^{k-1} explicitly rather than assuming it is a_0. Ordinary and exponential series mixed in one product. Partial fractions with a repeated root done as if the root were simple. A decomposition into parts that is not unique, so the product overcounts; the tidy-string decomposition was unambiguous because every block ends with the first non-A letter.

Ladder.

1. PS 11 #2. A binomial-type identity for falling factorials; prove it.
2. PS 11 #3. Subsets counted by size modulo three; a closed form.
3. PS 11 #7. Coefficients of a power of a short polynomial; a property of them.
4. 2017 B3. A power series with zero-one coefficients built from a recurrence; irrationality of its value.
5. 2023 A5. A signed sum indexed by digit counts; its complex roots.
6. 2020 B4. A weighted average over lattice paths; evaluate.
7. 2024 A6. A matrix built from the coefficients of a series; its determinant.

Where it hides. Inside recurrences and counting, as the method that does the bookkeeping (2020 A5, 2024 B5, PS 4 #12). Inside number theory, where coefficients are read modulo a prime or a power of a polynomial is expanded modulo 2 (2022 B1, PS 9 #2). Inside probability, where the generating function of a distribution gives its moments and the probability of a sum (2020 B4, PS 6 #12). Inside sums, where a sum of binomial coefficients with a pattern is a coefficient or a value of a known series (2020 A2). Inside linear algebra, where Hankel determinants of a series encode a continued fraction (2024 A6, PS 7 #24).